Jupyter Organization Tips

DATA SCIENCE BOOTCAMP

NASHVILLE SOFTWARE SCHOOL

Introduction

All of these slides are **recommendations**.

Goals:

- Reproducible Analysis
- Readable Code
- Notebooks which are Understandable by others (and your future self)

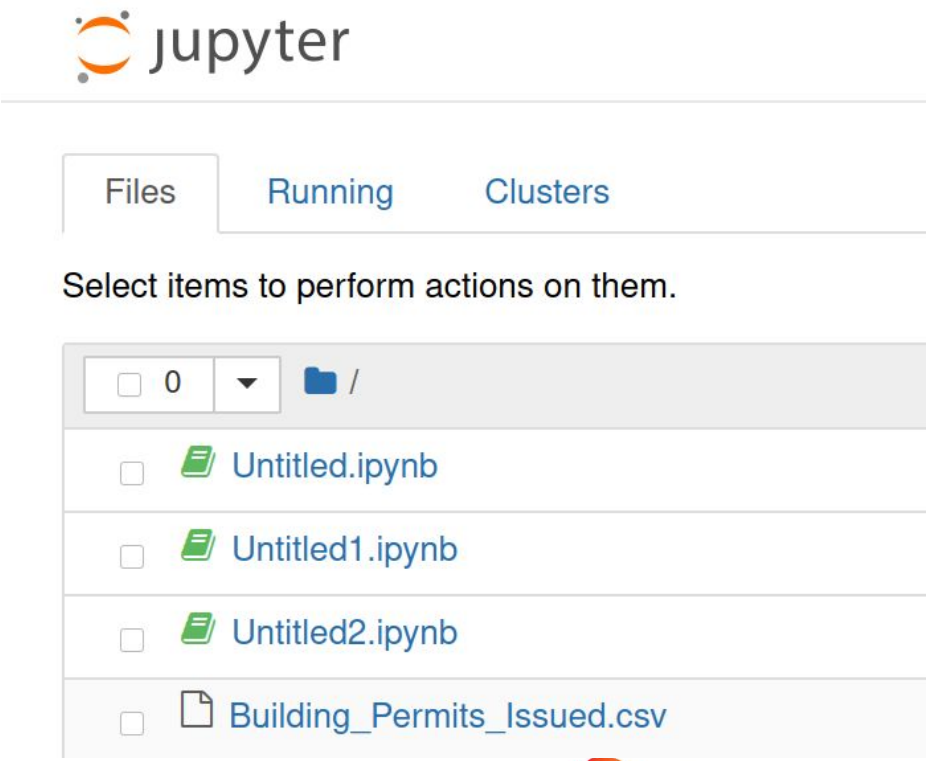
General Guidelines for Python Code

PEP 8 - Style Guide for Python Code

Guidelines to improve readability and consistency of code.

Recommendations

Give notebooks descriptive names!



Recommendations

Use Markdown to annotate your workflow.

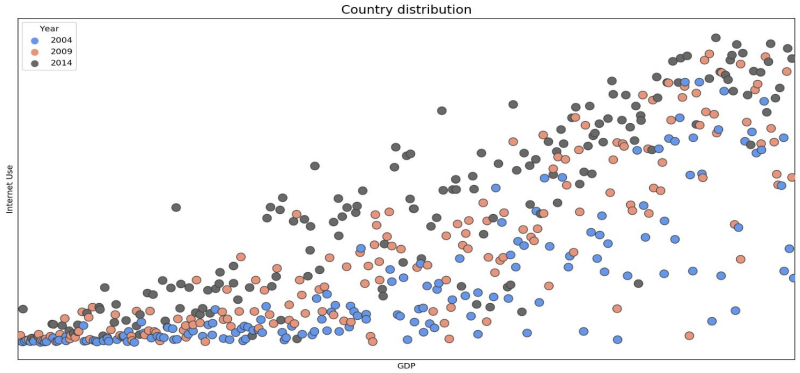
Comment code when appropriate.

Country GDP and internet usage distributions

Plotting of `Year` with x-axis as `GDP_Per_Capita` and y-axis as `Internet_Users_Pct`.

In [8]:
```
plt.figure(figsize=(16,8), clear=True);

ax1 = sns.stripplot(x="GDP_Per_Capita",
                    y="Internet_Users_Pct",
                    data=df_04_09_14,
                    jitter=2,
                    hue="Year",
                    size=10,
                    linewidth=.8,
                    dodge=True);
ax1.set_xlabel('GDP');
ax1.set_ylabel('Internet Use');
ax1.set_yticks([]);
ax1.set_xticks([]);
ax1.axes.set_title('Country distribution', fontsize=15);
```



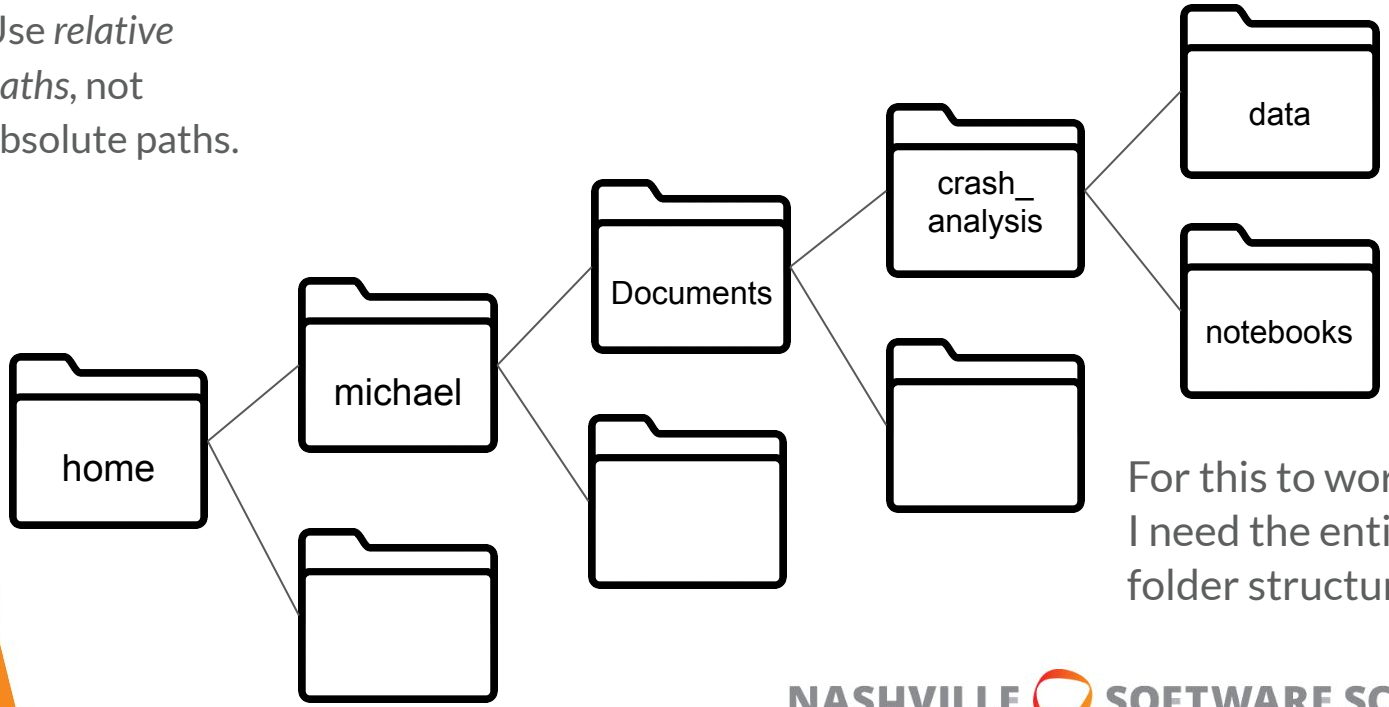
Observing the plot `ax1` above, we notice that in general, there looks to be a positive correlation between GDP and internet usage. This correlation seems strongest in years 2009 and 2014.

In [ ]:
```
# Data contains two lines of description text, skip to avoid errors.
df = pd.read_csv('data.csv', skiprows=2)
```

Recommendations

```
crashes = pd.read_csv('/home/michael/Documents/crash_analysis/data/crashes_2018.csv')
```

Use *relative paths*, not absolute paths.

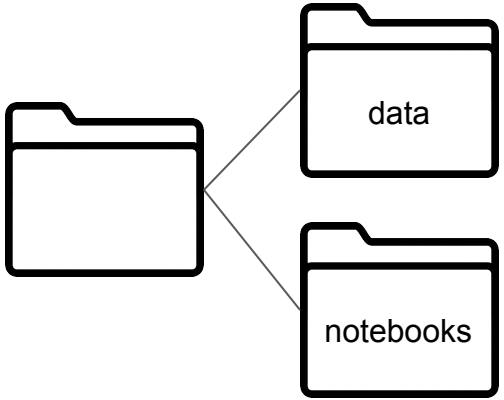


For this to work, I need the entire folder structure!

Recommendations

```
crashes = pd.read_csv('../data/crashes_2018.csv')
```

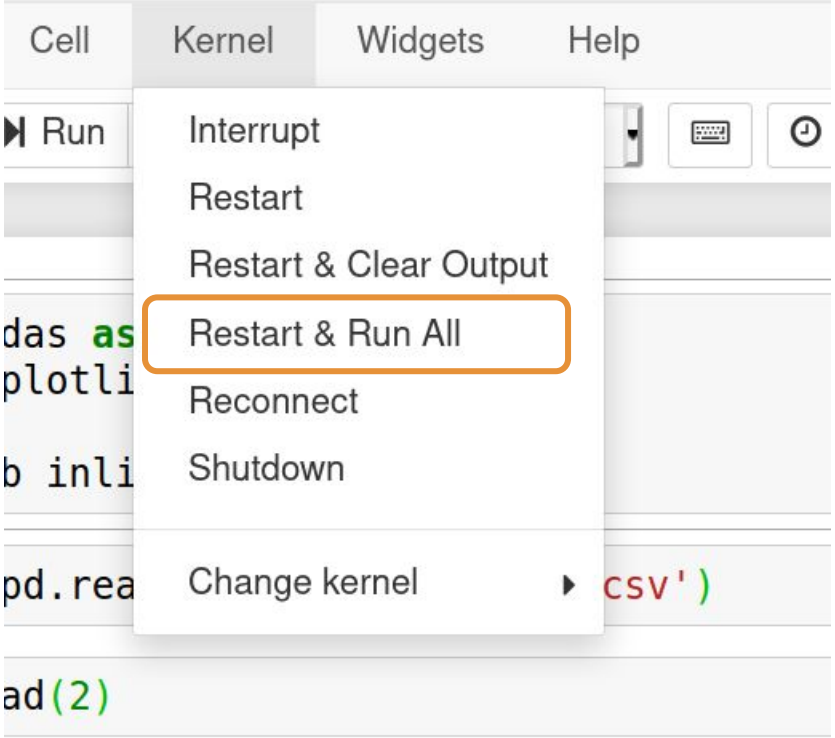
Use *relative paths*, not absolute paths.



For this to work, I only need the folder structure within the project folder.

Recommendations

Periodically check that you can Restart & Run All error-free.



Recommendations

Use logical and human-readable variable names.

```
df = pd.read_csv('Building_Permits_Issued.csv',
                 low_memory = True)
```

```
df2 = df[df['Permit Subtype Description'].
         str.contains('Residential')]
```

```
building_permits = pd.read_csv('Building_Permits_Issued.csv',
                               low_memory = True)
```

```
residential_permits = building_permits[building_permits['Permit Subtype Description'].
                                       str.contains('Residential')]
```

Recommendations

Avoid inplace = True when using pandas methods.

When making a change to a DataFrame, explicitly assign the result back. This makes it obvious to someone reading your code that you are modifying the DataFrame.

```
derby.drop(columns = 'starters', inplace = True)
```

```
derby = derby.drop(columns = 'starters')
```

Recommendations

Combine together multiple cells if they don't produce output

Take advantage of method chaining.

```
In [4]: crashes = crashes.drop(columns = ['Weather Code', 'Illumination Code', 'Harmful Code', 'Mapped Location'])

In [5]: crashes = crashes.rename(columns = lambda col: col.lower().replace(' ', '_'))

In [6]: crashes = crashes.dropna(subset = ['latitude', 'longitude'])
```

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        crashes = crashes.rename(columns = lambda col: col.lower().replace(' ', '_'))
        crashes = crashes.dropna(subset = ['latitude', 'longitude'])
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Recommendations

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In [5]: crashes = crashes.rename(columns = lambda col: col.lower().replace(' ', '_'))

In [6]: crashes = crashes.dropna(subset = ['latitude', 'longitude'])
```

```
crashes = (
    crashes
    .drop(columns=["Weather Code", "Illumination Code", "Harmful Code", "Mapped Location"])
    .rename(columns=lambda col: col.lower().replace(" ", "_"))
    .dropna(subset=["latitude", "longitude"])
)
```